

Chapter 14

Theoretical and Methodological Innovative Futures for Physics in Postsecondary Contexts



Eleanor S. Armstrong, Allison J. Gonsalves, Anna T. Danielsson,
and Emma Mckay

Abstract The aim of the chapter is to highlight promising theoretical and methodological innovations that work to understand and address inequities in postsecondary physics contexts. In particular, we move beyond a focus on the individual and identity as the main lens. Instead, our focus on relationships pulls a range of new research sites into consideration as vital sites of postsecondary physics education, where new, existing, and non-members of the field learn to navigate the profession. We show how relations between people, communities, and institutions can unsettle what is focused on in physics education research (PER).

Keywords Physics education research · Postsecondary physics · Communities · Theoretical innovations · Methodological innovations

14.1 Introduction and Overview

The aim of the chapter is to highlight promising theoretical and methodological innovations that work to understand and address inequities in postsecondary physics contexts. In particular, we move beyond a focus on the individual and identity as the

E. S. Armstrong (✉)

Institute for Space Research, University of Leicester, Leicester, United Kingdom

e-mail: ea377@leicester.ac.uk

A. J. Gonsalves · E. McKay

McGill University, Montreal, QC, Canada

e-mail: allison.gonsalves@mcgill.ca; emma.m.mckay@mail.mcgill.ca

A. T. Danielsson

Stockholm University, Stockholm, Sweden

e-mail: anna.t.danielsson@su.se

main lens. We instead show how a focus on relations between people, communities, and institutions can unsettle physics education research (PER) and offer new avenues for scholarly investigation. In particular, if PER researches education, then its hegemonic contexts, interests, goals, and methods play a role in defining what education *means*. The broad overview we provide in this chapter of questions and re-orientations for all of these pieces of PER pushes then toward reconceptualizing what counts as education within PER, moving toward relationships, community, and critical critique as core to the idea of education.

This chapter takes up and responds to two important contexts. First, PER and physics equity work more broadly has primarily been conducted in ways which support seeing science education as a process in service of developing capital (see, for example, Pierce, 2012) and, consequently, have focused on “diversity” and “inclusion” issues like binarized gender inequalities. Second and more specifically, PER has traditionally focused on a genre of research that prioritizes learning standard physics knowledge and practices in the university classroom and student laboratories—a context which has only recently started to challenge the colonial, enlightenment values of physics research at large, instead continuing to train and socialize individuals into these practices. In this chapter, then, we focus on how attending to efforts that grapple with “politicized” questions and themes, and the social backgrounds to the research in the community, can broaden the field of (postsecondary) physics education research. We look at examples of action and learning in a variety of contexts, which responds to the call to pluralize the typical focus of PER on classroom contexts.

14.1.1 Positionality

As researchers, we acknowledge that research is a subjective endeavor that is situated in our varied positionalities. All four authors bring varying relations with ancestors, the lands we live on, and physics as a subject, to our scholarship on physics education. We recognize that we have White racialized identities, and are/have always been settlers; all four of us living and working in colonial states. We understand that engaging in research means also recognizing those privileges and colonized perspectives in our research and work. Our epistemological positionalities are aligned with research that seeks to identify and challenge issues of power, equity, and social justice in physics contexts, particularly in academia. Some of us position ourselves as insiders to physics, while others are outsiders to physics but benefit from positions as insiders to postsecondary physics.

14.1.2 Research Context for Postsecondary Physics Education

Often motivating physics education research is the goal to have more (and more diverse) people in STEM. Why do we need more people working in STEM? An answer to that question involves complex, entangling ideas about scientists and engineers as contributing to economic growth with considerations about the significance of STEM professionals for a country's defense capabilities and national security (Fensham, 1985; Lövheim, 2016, 2021). To complicate matters further, utilitarian arguments concerning a perceived need for more STEM professionals are often interspersed with arguments drawing on equity and social justice (DeBoer, 2013). This emphasis on "getting more qualified professionals into STEM fields" has led to research on barriers that under-represented groups face in STEM, and solutions to support their trajectories into the "STEM pipeline," rather than questions about the STEM pipeline itself. In physics education research in particular, the issue of the uneven distribution of women and men in physical sciences has received considerable attention. Looking back, notable works include Laura McCullough's research on gender differences in physics assessments (e.g., McCullough, 2004), the work on highlighting and reducing gender gaps in higher education physics (e.g., Pollock et al., 2007), and research in secondary schools investigating "physics identity" and strategies to encourage girls to enter university-level physics programs (e.g., Hazari et al., 2013). Such work has been pivotal in uncovering binarized gender inequities in physics, and in postsecondary physics education especially.

Casting the net wider, feminist critiques of science (e.g., Fox, 1985; Prescod-Weinstein, 2020) and studies of physics communities from anthropological points of view (e.g., Traweek, 1988; Hasse, 2002) have provided complementary perspectives on the intersection between gender and the epistemology and practices of physics. This contrasts to much equity work in physics education oriented work in the context of postsecondary physics that has, implicitly or explicitly, relied on notions of gender as something static and binary (Traxler et al., 2016). Lately, this work has diversified to include both a recognition of a wider set of social categories (e.g., Barthelemy's (2020) work on LGBT+ physicists) and expanding theoretical vantage points, such as critical race theory (e.g., Hyater-Adams et al., 2018) and intersectionality (e.g., Avraamidou, 2020a). Such identity-oriented perspectives have lately gained a lot of traction in physics education research (Gonsalves & Danielsson, 2020) for understanding connections between gender and who is recognized as a particular kind of physics student or physicist. As useful as this work has been in uncovering nuances of exclusionary practices in physics, we cannot but wonder where the perspectives' limits are in terms of reimagining teaching, learning, and research practices in physics. Therefore, in this chapter, rather than give an overview of equity-oriented research in postsecondary physics education research to date, we look forward toward future possibilities. We seek to identify strands in the contemporary research that we see as auspicious for engaging a diversity of learners and disrupting harmful norms and practices in postsecondary physics contexts.

14.1.3 Grounding Lens and Showcasing Innovations

In this chapter, we cover three innovations: theoretical innovations; methods and practices; and rethinking impact. In these sections, we draw into PER context developments that have taken place elsewhere in academic scholarship; and demonstrate their use and future potential uses within postsecondary physics DBER. We have selected some examples from different frameworks which are themselves expansive, and we recommend that the references we include here can act as a guide for those interested in reading and thinking with these theoretical frameworks at large.

Our view of postsecondary contexts will include not just higher education programs and university students' experiences in physics, but also learning in professional communities, informal learning spaces in postsecondary physics institutions, and community engaged physics research. Learning and education have long been conceived of expansively (e.g., Sandlin et al., 2011) to include extra-institutional contexts; de-institutionalizing the conception of teaching and learning and liberating knowledge development. In expanding our focus beyond traditional learning institutions, we draw inspiration from Herrenkohl et al. (2019) who argue for a "learning humanities" that puts relationships at the center of inquiry. Thus, we move away from examining the cognitive dimensions of physics learning, and from the examination of practices shaping identity work, to instead explore how relationships are centered in (in)formal physics practices, and how practices which place relationships at the center can transform forms of engagement in physics. Thus, we no longer wish to answer questions about how to get more qualified people into physics. Instead, we seek to learn from research and practices that ask questions about obligations vis-a-vis physics practices, responsibilities that physicists and physics students have to act in the world (in relation to human and more-than-human actors), and mechanisms that retain those who are qualified in the profession. Our focus on relationships pulls informal learning contexts into consideration as vital sites of postsecondary physics education, where new, existing, and non-members of the field learn to navigate the profession.

Thus, In Sect. 14.2, we cover theoretical innovations that engage with critical race and critical Whiteness theories and queer theory, as well as exploring expansively other theoretical innovations elsewhere in science learning contexts that we see innovative futures in for postsecondary physics education research. In Sect. 14.3, we demonstrate how mentorship communities and professional development networks offer novel methods and practices toward futures in postsecondary physics communities that are not structured simply around "including" new folks into existing communities of practice. Finally, in Sect. 14.4, we pull to the fore how innovation in what university outreach is and how it is studied in PER contexts, and in the practices and research of citizen science exercise can expand postsecondary physics education. We conclude the chapter looking forward to innovation in futures in postsecondary PER.

14.2 Theoretical Innovations Adopted Recently in PER

In this section, we show how innovations in application of theoretical lenses offer critical perspectives on PER and innovative futures toward which we can expand physics education (and the research that studies these practices expansively).

14.2.1 *Critical Race Theory and Critical Whiteness*

Recent developments in postsecondary PER have seen an uptake of critical perspectives on race in physics education, drawing on critical race theory, intersectionality, and critical Whiteness (Rosa & Mensah, 2016; Hyater-Adams et al., 2019; Avraamidou, 2020b, 2022; Robertson & Hairston, 2022). Critical race theory (CRT) describes racism as an everyday ordinary occurrence, understanding that race structures social interactions, and that racism is normalized in social systems, social institutions, and social norms. This normalization of racism is an important principle that CRT seeks to make visible as a structuring feature of society (Delgado, 1995). As race, through the lens of CRT, is produced in social interactions, to understand racism research must center the voices of People of Color (Delgado & Stefancic, 2017). Rosa and Mensah (2016) documented the voices of Black women in physics, by utilizing one of the key principles of CRT: counterstories. In centering the stories of Black women in physics, telling counterstories confronts the dominant stories of those who are in power in physics. These counterstories revealed obstacles that Black women face in physics, such as ordinary racism and microaggressions, and frames these obstacles, such as feelings of isolation and decreased self-efficacy, mechanisms of exclusion which compromised their learning and performance in physics. Rosa and Mensah argue that traditional narratives in research about persistence in physics overplays “academic achievement...and downplays the role of the racial social climate in STEM fields” (p. 11).

The counterstories enabled by the CRT lens reveal how racism functions specifically in students’ performance and persistence in physics, and challenges the dominant narratives that place student deficits as a key feature that interventions in the field should seek to address. In this study, Hyater-Adams et al. (2019) adopt a CRT lens to investigate how Black folks identify as physicists, developing a framework (critical physics identity, CPI) to describe such identities. The framework draws on the basic tenets of science identity framings (featuring the importance of recognition, performance, and competence) but also combines these with Nasir’s (2011) model of racialized identities that foregrounds the importance of relational, ideational, and material resources in the construction of identity. The findings showed that all participants strongly identified with physics, but experienced difficulties performing their physics identities due to similar structural barriers as those

described by Rosa and Mensah. The value of this framework which places race at the analytical centre is that it provides a means to capture the counterstories of Black participants, while identifying the structural obstacles that impact Black students' performances in physics.

Drawing on intersectionality theory, Avraamidou (2020b, 2022) challenges the prioritizing of the "politics of recognition" in physics identity research, asking researchers to consider where the power resides in research that seeks to learn about how recognition supports insider identity work in physics. Avraamidou's use of intersectionality—which develops an understanding of how the way people experience society is shaped by the overlap of their different identities that are inextricably interlinked and not separable (Nash, 2011)—in this research highlights the often problematic ways that the concept of recognition is mobilized in identity research (in PER and science education research more broadly). Recognition has emerged as a strong analytical concept in identity research investigating the underrepresentation of women and racialized people in physics (e.g., Hazari et al., 2010). However, Avraamidou (2022) points out, recognition has been largely mobilized as a binary concept (either one is recognized as a physics person, or not), which neglects the multiple, intersecting ways that identity positions can yield various forms of recognition, non-recognition, or misrecognition that are not fixed between different contexts. Thus, Avraamidou argues, there is a need to attend to the politics of recognition in identity research, and especially to attend to the ways that current conceptualizations of recognition "leaves open the possibility of women in physics seeking recognition from the very people and institutions that are responsible for their underrepresentation" (p. 62).

Robertson and Hairston (2022) operationalize Critical Whiteness theory to examine how White supremacist ideology can structure interactions in physics classrooms. The authors draw on CRT and literature from Critical Whiteness studies to identify "markers for whiteness" and to analyze how these show up in introductory physics classrooms. This research examines micro-interactions between students, a teacher, a whiteboard, and a physics concept to analyze the interactions and the teacher's attention. Mobilizing several concepts from CWT, the authors demonstrate how interactions in the classroom center Whiteness in a way that elevates dominant voices, ideas, and ideologies while marginalizing others. These interactions, and those described in follow-up interviews about the class, demonstrate how meritocracy is intertwined with Whiteness in ways that are valued among students and the teacher. These ideologies are likely to be silencing to many of those participating in the class, yet they are reproduced by students in interactions. This new research in physics education is a radical departure for the field. The response to its publication signals the troubling relationship physics education has with race and racial ideology. Shortly after this paper's publication, the editors of *Physical Review Physics Education* journal published an editorial supporting the research and their choice to publish it, and admonishing the "harassing, intimidating, and diminishing tweets and emails in response to the publication of the article" (Henderson & Thoennessen, 2022). The authors have since published subsequent work employing a CWT framework that demonstrates how faculty

members teaching in mainstream physics programs will adopt “race evasive frames” (e.g., grit or equal opportunity) to discuss representation and success in ways that maintain race-neutrality and the minimization of racism in physics (Robertson et al., 2023).

14.2.2 *LGBT+ Research and Queer Theorising*

Where early equity and diversity work in PER was almost exclusively focused on binarized gender, recent work has instead expanded to consider the experiences of LGBT+¹ physicists and the constructions of gender and sexuality. Rather than only accounting for the experience of cis women and cis men, this research reports on the experiences of trans and non-binary physicists. A notable series of papers by Ramón S Barthelemy and colleagues (Barthelemy, 2020; Barthelemy et al., 2022a, 2022b) report from an online quantitative and qualitative survey of LGBTQ+ physicists, exploring LGBTQ+ physicists and physics students experiences of exclusionary behavior and harassment. While those respondents who could be out about their LGBTQ+ identity were more comfortable with their institution (Barthelemy et al., 2022a), in the 71 qualitative responses that reported on experiences of exclusionary behavior and harassment for cis and trans physicists, Barthelemy (2020) found that the majority were based on non-normative gender expression, including for cis physicists, and being a woman. Their data demonstrate that nearly half of the transgender respondents experienced the most exclusionary behavior (with one fifth reporting at least one experience of this in the past year), and over one third of transgender respondents considering leaving their workplace. This finding is echoed in a survey of astronomy and planetary scientists, where Richey et al. (2019) found that although LGBTQPAN women and gender minorities were more likely to feel unsafe at work due to their race, sexual orientation, and gender identity than cisgender, straight women.

To date, this line of research has focused on the experiences of LGBT+ students and scientists, and contributes to broadening the discussion about diversity and inclusion in PER beyond binary gender; demonstrating the need to take into account the intersection of different axes of power. There have been, for example, vital calls to adapt survey practices and institutional policies to take a complex approach to gender, specifically moving beyond gender as something binary, discrete, or that can be easily discernible through names or other external characteristics (Rasmussen et al., 2019; Strauss et al., 2020). Similarly, raising the profile in PER projects of both LGBTQ+ physics students and physicists through developing resources about prominent LGBTQ+ folks (e.g., 500 Queer Scientists, n.d.), or of popular culture

¹We use the term LGBT+ in our writing, but where we refer to findings from other papers, we use the language in the original text. This is to honor the ways that the researchers have worked to categorize their participant experiences, and highlights the plurality of approaches in the scholarship to date.

figures engaging with physics such as Amrou Al-Kadhi describing quantum mechanics' non-binary state as "beautiful, strange and glorious" (Al-Kadhi, 2024) has been a vital step toward developing both representation and identity-confirmation in postsecondary physics education.

However, we might distinguish this work on LGBTQ+ visibility and recognition, to practices that embed queer theory and queer approaches to both PER and postsecondary physics education contexts. In postsecondary informal science education contexts, Armstrong (2022) has demonstrated the use of queer theorizing to unpack "hidden" narratives that both discuss LGBTQ+ identities and moved to apply critical queer perspectives on physics and other science research for tour guide attendees in a museum. This demonstrates the potential of queer theorizing in postsecondary physics education where it seeks to deconstruct dominant narratives around gender and sexuality, dismantling binary constructions such as heterosexual/homosexual and masculine/feminine. Elsewhere, work to theorize queer epistemologies of identities in physics contexts is taking shape (see, for example, Armstrong & Danielsson, 2023, and Orthia & Roberson (2023)). In a recent anthology, Letts and Fifield (2019), who have collected STEM education writings engaging with queer theories over two decades, state that "Queer theorizing uncovers and incites paradoxical relations of the normal and strange in ways of knowing, feeling, and being ...[and attends to] processes that normalize and naturalize (and abnormalize and denaturalize) particular manifestations of human subjects, society, culture, and nature." (p. 4). This can be seen in Helene Götschel's (2019) physics university classroom. She seeks to teach queering of physics by, for example, "scrutinizing what is supposed to be normal and what is invisible or silenced, making the familiar strange, reflecting on the discursive production of physical knowledge" (p. 126). In this way, Götschel also discusses how our understanding and storying of the world through physics sometimes relies on binary, heteronormative gender systems; reflective of Barad's work to use queer to rethink the very ontology of physics (Barad, 2011).

14.2.3 Theories Used More Widely in Other Science Learning Contexts

In addition to these two theoretical vantage points, we note that there are a range of innovations that have been taken up elsewhere in both non-physics postsecondary science education and physics education at other levels, and here we gesture toward the possibilities for postsecondary physics education research by engaging with these rich bodies of work. Collectively these (along with the queer and critical race theory and Whiteness sections) share a willingness to look at physics as a system of knowledge that is part of larger social systems and to critically to engage with social justice in the context of physics education (Cochran et al., 2021).

Well integrated into research on early-years learning contexts, posthumanism and other theories which include matter offer theoretical innovation for postsecondary PER by emphasizing the co-emergence of learning between learners and matter. Introducing this into theorizing of postsecondary offers mechanisms to engage with how, in contexts beyond the classroom, engagement and experience are often a priority. This is in direct contrast to the formal classroom where “often concepts are afforded more ontological power and are considered more real than the lifeworld experiences of the learner” (Milne, 2019). Particularly rich in theorizing around elements of the natural environment, for example, in exploration of water by Elkin Postila (2022), or engaging learners in ethical education in science (e.g., Scantlebury & Milne, 2020), these approaches offer the potential to draw attention to dimensions of physics education practice that are currently under-theorized in the postsecondary domain. Such work directs our attention to the formation of identity through intra-action with matter (Barad, 2007), and in the social engagement with the nonhuman world. However, an emphasis on the genealogies of posthumanism as *the* mode of engaging with the interaction of human and non-human worlds in other science learning overlooks and erases theorizing by Indigenous scholars and with Indigenous knowers (Todd, 2016). For example, the Turtle Island Institute (founded by Melanie Goodchild (Goodchild 2023)) supports a Virtual Teaching Lodge that engages in Indigenous-led teaching of science specifically beyond the university classroom; Aboriginal and Torres Strait Islander Astronomy project works with teacher education in the postsecondary context at the University of Melbourne (*Aboriginal and Torres Strait Islander Astronomy*, 2022).

Agitating for greater engagement in the implications of physics being part of a settler colonial project in the education that is provided about physics is part of activist work within the field (see Sect. 14.3: Methods and Practices for more details). Deliberate engagement with, and challenge to, entitlement to Indigenous land and access for physics research can be seen in relation to large astronomical science operations (e.g., Casumbal-Salazar, 2017; Neilson, 2019; Goodyear-Ka’ōpua, 2020; Maile, 2021). But, thus far, the impacts of including such critiques in physics education, for example, in the content of classroom work or ethical skills for engaging communities of practice, has been undertheorized in PER. Such work could unpack what work postsecondary learning does or could do to (re)shape beliefs that physics as a body of knowledge is modern, contemporary, progressive, future-oriented *and* that those attributes are good, desirable, and/or inevitable; highlighting where or how learners both within and the beyond the field are socialized to believe this to be the case. In this way, we call for innovative futures in physics education that move research past the support of colonial states and assimilatory equity that currently dominates the field. All of these critiques encourage us to ask: What will it take in PER to interrogate how colonial, capitalist, and extractivist practices, systems, and theories exist in physics?

14.3 Methods and Practices of Physics Education

We highlight postsecondary contexts that expand the concept of “physics education” to include learning beyond teaching in the classroom. Our contribution here is to move a focus on methods and practices of physics education in the university classroom context to more varied social contexts. We exemplify by one practice in the context of under-graduate students and one in the context of professional learning, but recognize that these practices also are powerful across contexts and that the distinction between “professional” and “student” is porous and not clearly defined.

14.3.1 *Physics Learners: Mentorship and Training Programs*

Socialization into the field of physics has been an important and long-standing focus of research in PER (e.g., Danielsson, 2012; Gonsalves, 2014), theorizing at how learners learn to belong in the research field. Socialization takes place in many venues. Beyond learning through university classrooms, mentorship and training programs can provide such socialisation events. A wave of programs focus on providing opportunities for more inclusive socialization for those who come from minoritized backgrounds into the field. However, even well-intentioned mentorship provided by people outside the community from which the junior participants come from, and especially that undertaken by White scholars, can potentially be alienating. Counter-narrative led research by Martinez-Cola (2020) demonstrates that Students of Color are subject to imbalance mentoring relationships that, especially in their worst cases, do not see challenging of community norms in developing a place within the community for the mentees, instead trying to fit those outside into existing norms. While there are some White mentors who act as “Allies” (as Martinez-Cola describes them) to disrupt community norms through inviting mentees to events, introducing them to their community, co-authoring with them, she makes the point that these are infrequent persons in the academy. Rather, many mentors “collect” students (then extractively utilize mentees for personal gain) or “nightlight” mentees (by shining guiding lights for mentees without recognizing systemic challenges) in mentoring relationships which socialize students in harmful ways.

To support physics learners, developing communities can be vital. Community change can be supported, convened, led, and populated by the minoritized groups that are historically excluded. These projects offer a different idea about what it means to be socialized into the field, in what we might call “by-with-for PER projects,” led by community members, for community members, with community members as active agents in their development (an adaptation of Simon, 2016). For example, at the California Academy of Sciences, a summer intern program for queer youth, “STARS” (Striving for Trans Inclusion and Anti-Racism in Science Learning) lead by trans and/or BIPOC museum workers to forge paths that see queer identities

as part of postsecondary science fields. As an intern on the program, Eli Ramos (2021) gathered the experiences of their colleagues on this program. Ramos notes that many colleagues were united in their “ostracization by race or gender” from science community spaces prior to the program. However, STARS interns described how development “alongside other people of color and LGBTQ+ people has been instrumental to us” allowing them not to have “to explain the basics of my identity and justifying my personhood”—which “was a relief” (Ramos, 2021). Elsewhere, participants at Space Science in Context reported their experience of not having to explain or justify themselves in a conference setting (based on racialized, ableist, or misogynistic judgements) as being integral to finding a community in space sciences that encouraged them to see they belonged in the field (Persaud & Armstrong, 2022). Such experiences describe the relief of not having to do what Murray et al. have theorized as the additional “imposter work” that minoritized folks often undertake to survive and thrive in postsecondary science education contexts (Murray et al., 2022) by being outside the social conditions that make them outsiders to the science community at large. This shift in focus from the individual to the system of relations that maintain the internalization of “feeling like a fraud despite one’s achievements” (p. 7) rejects the need for individualized success narratives that often belie the privileges that are required to experience overcoming such exclusions. As Hodari et al. (2022) urge in their paper on socialization in physics training it “costs us so little to counter those experiences with cues that tell them we are happy they [minoritized students] are there” (p. 21).

14.3.2 Physics Professionals and Communities: Reshaping Professionalism Through Community Building and Action

Other postsecondary physics learning spaces (re)develop norms within professional STEM communities, which are by members of the profession, for their peers, in community with them. Although the distinction is somewhat arbitrary from the section (what does “to be within the field” mean as opposed to becoming part of the field, for instance), in this section, we focus on professional development and community building projects and programs. In their paper, Hodari et al. (2022) note while much research focuses on undergraduate contexts, late-postgraduate and postdoctoral positions in physics fields are key times where researchers are socialized into the field. It is during these periods that “historically marginalized scientists are often excluded or kept out of these professional networks that are tied to increased productivity, which in turn affects their professional success” (p. 4).

Thus, the development of space to reshape social norms of the field from within are vital, which can lead to relearning what participation within the professional physics field could be. These take a wide range of guises from labs within institutions, to trans-national digital communities. From digital community building of

projects such as #BlackinSTEM, Particles for Justice (n.d.), Pride in STEM (n.d.), Shutdown STEM (n.d.), VanguardSTEM (2021) leading to off-line action, to projects such as the *Astronomers for Planet Earth* which encourages climate conscious research and conference participation (Burtscher et al., 2022); or to local institutional LGBTQ+ STEM community action across science departments at UCL (Lock et al., 2025); it is possible to expand the focus on professional community change. We unite these projects in their work to support relearning behaviors within the postsecondary physics community, through supporting, validating, and focusing change in the community and institutions at large through reshaping ideas about how physicists can be within the community. In their work on the experiences on Women of Color in STEM, Ong et al. (2018) describe how such communities can be theorized as counter spaces for Women of Color, instrumental for “facilitating their educational preparation and readying them for careers and beyond” (p. 216).

14.4 Rethinking the Impacts of Informal Postsecondary Learning Contexts

While existing literature in the field addresses the impact of informal learning primarily on the participants in such projects who are from “the public,” we here make an intervention to draw attention to the work that is being done by those developing informal learning practices and the opportunities that this provides to reshape their understandings and knowledge of physics. We focus particularly on reconceptualizing *who* is learning the practices through the example of University outreach and *what* is being learned through exploring innovative citizen science projects.

14.4.1 University “Outreach”

Engagement with informal learning in postsecondary physics contexts has been shown to impact university students’ identity work in their physics programs in ways which expand the sense of what physics identity is or ought to be. These programs can be meaningful opportunities for students to build community and find a sense of belonging within their university programs. Research investigating the impacts of university students’ experiences as facilitators of physics outreach projects suggests that informal physics learning experiences can have positive impacts on students’ identity work outside of the constraints that are usually found in academic physics settings (Fracchiolla et al., 2020). Participating in a physics outreach community of practice provides opportunities for students to be recognized as members in the physics community in new ways, and expanding their participation beyond the normative practices of university physics (Prefontaine et al., 2021). This is not limited to university students. Outreach practices of researchers also provide new spaces to negotiate their science identities through creating and sustaining

multiple science identities theorized as giving researchers the opportunity to experiment with developing shifting and contingent identity performances in university science outreach (Davies, 2021). Additionally, research specifically investigating women's involvement in facilitating informal physics programs found that participants gained increased opportunities for internal and external recognition, thus providing opportunities for them to "curate their own sense of expertise" which improved their confidence in their chosen major (Randolph et al., 2021, p. 363).

These programs also provide resources for youth who are engaged in learning that facilitates their own socialization into physics. For example, research into the significance of "university adjacent" informal physics learning programs suggests that university outreach programs provide numerous resources that contribute to youths'—especially young women's—identity work vis-a-vis physics (Gonsalves et al., 2022). Programs identified as university-adjacent (such as outreach programs, internships, summer schools, and public lectures) can provide opportunities for youth to begin to see themselves as physics people, and to learn about the lives of physicists beyond the classroom or even the lab. While we acknowledge the importance of these practices to contribute to pluralizing the people who participate in physics, we also wish to point to these as often elite and difficult to access programs for many minoritized students.

14.4.2 Citizen Science

"Citizen science" designates a wide range of different activities involving public participation in science. This can include, for example, amateur naturalists collecting observational data about birds (Parrish et al., 2019), people classifying images of galaxies (Reed et al., 2013; Lingard et al., 2020), or monitoring of radioactivity in food (Reiher, 2016; Andersson-Sundén et al., 2019). Today, citizen science is typified by distributed fieldwork or crowd-sourcing of data analysis (McQuillan, 2014), with the common goal of "educating" the public about science. Research in PER related to these activities have hegemonically, thus, looked at the "success" of various projects against such goals. But citizen science has an epistemically transformative potential: challenging the very foundation of the modern regime of scientific knowledge production where the work of professional scientists is separated from the lay public, who are understood as—at best—consumers of science. The involvement of lay people in science not only changes who participates in the knowledge production, it can also transform how this is done as well as where and why—including lay people in experimental research not just in collection of observations (Strasser et al., 2019). Strasser and colleagues instead attend to "striking" importance of the perceptive and affective qualities employed by participants during citizen science projects, despite "knowing that [these attributes'] exclusion was a key element in the formation of modern science, based on the ideal of objective, rational, and disinterested knowledge" (2019, p. 65).

McQuillan (2014) takes this potential further, arguing for citizen science as nomadic science focused on the local, the situated, and the culturally reflective. In this nomadic science also lies a potential for community activism, to enact science as a bottom-up practice, attentive to local needs and practices (McQuillan, 2014). For example, the Public Lab (n.d.) seeks to pursue environmental justice through community science and open technology. Through low-cost tools, people are able to participate in the building of community knowledge about environmental health based on physics experimental techniques (e.g., do-it-yourself spectrometers used to detect oil pollution). Another example of citizen science within physics is the Original Research By Young Twinkle Students (ORBYTS) program, where secondary school students have been involved in original research on exoplanets, supervised by PhD students and postdocs (Dunn, 2021). Part of the data used comes from data obtained by citizen astronomers, thereby linking junior research, school students, and citizen scientists. One aim with the ORBYTS program is to by involving school students in state-of-the-art research empower them to participate in STEM and challenge ideas about who can be a producer of scientific knowledge (Sousa-Silva et al., 2017).

14.5 Conclusions, Contributions to the Field, and a Call to Innovative Futures

Throughout this chapter, one of our overarching contributions has been to (re)conceptualize what counts as *education* within the field of physics education research by introducing and highlighting theories, methods, and contexts that are a departure from hegemonic PER. In our selection of cases, we particularly ask what it means to think of *education* beyond the normative postsecondary contexts of the university classroom, with a greater focus on *learning* in its expansiveness to capture experiences, knowledge transfers, skills, and behaviors. Not only does this transformation open a range of new practices and sites of subjectivities to the purview of PER, it also moves beyond a conceptualisation of postsecondary education that is narrowly focused on those for whom physics remains a central part of their lives. Instead, this view allows exploration of the value of understanding what physics learning might mean for those beyond the field. Thus, we call for innovative moves in postsecondary PER: from the individual to networks of relationships, from a specifically university classroom context to more varied social contexts, and from support of socializing people into physics more efficiently into critiquing colonial physics at its core. This transformation of physics education research and practices is a call to think critically not only about the sites of physics learning that are researched, but also the goals of the research practice. We call for innovative futures in PER that resist reproducing oppressive, colonialist, extractive cultures, or the goals of improving “diversity and inclusion” in physics, for example, and instead invest in imagining sociocultural change beyond increasing representation.

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